```
#include <stdio.h>

int main() {
    int n, m, i, j, k;
    printf("1WA24CS216\n");

    printf("Enter the number of processes: ");
    scanf("%d", &n);

    printf("Enter the number of resources: ");
    scanf("%d", &m);

    int allocation[n][m];
    int request[n][m];
    int available[m];

    int work[m];
    int finish[n];
    int safeSequence[n];

    printf("Enter the allocation matrix:\n");
    for(i = 0; i < n; i++) {
        printf("Process %d: ", i);
        for(j = 0; j < m; j++) {
            scanf("%d", &allocation[i][j]);
        }
    }
```

```
printf("Enter the request matrix:\n");
for(i = 0; i < n; i++) {
    printf("Process %d: ", i);
    for(j = 0; j < m; j++) {
        scanf("%d", &request[i][j]);
    }
}

printf("Enter the available resources: ");
for(i = 0; i < m; i++) {
    scanf("%d", &available[i]);
    work[i] = available[i];
}

for(i = 0; i < n; i++) {
    int allZero = 1;

    for(j = 0; j < m; j++) {
        if(allocation[i][j] != 0) {
            allZero = 0;
            break;
        }
    }

    if(allZero)
        finish[i] = 1;
    else
        finish[i] = 0;
}

int count = 0;
```

```
while(count < n) {
    int found = 0;

    for(i = 0; i < n; i++) {

        if(finish[i] == 0) {

            int possible = 1;

            for(j = 0; j < m; j++) {
                if(request[i][j] > work[j]) {
                    possible = 0;
                    break;
                }
            }

            if(possible) {

                for(k = 0; k < m; k++) {
                    work[k] += allocation[i][k];
                }

                safeSequence[count] = i;
                count++;
                finish[i] = 1;
                found = 1;
            }
        }
    }
```

```
    if(found == 0)
        break;
}

int deadlock = 0;

for(i = 0; i < n; i++) {
    if(finish[i] == 0) {
        deadlock = 1;
        break;
    }
}

if(deadlock) {
    printf("\nSystem is in DEADLOCK state.\n");
    printf("Deadlocked processes are: ");

    for(i = 0; i < n; i++) {
        if(finish[i] == 0)
            printf("P%d ", i);
    }

    printf("\n");
}
else {
    printf("\nSystem is in safe state.\n");
    printf("Safe Sequence is: ");

    for(i = 0; i < n; i++) {
        printf("P%d ", safeSequence[i]);
    }
```

```
        printf("\n");
    }

    return 0;
}
```

"C:\Users\Admin\Desktop\1W

```
1WA24CS216
Enter the number of processes: 5
Enter the number of resources: 3
Enter the allocation matrix:
Process 0: 0 1 0
Process 1: 2 0 0
Process 2: 3 0 3
Process 3: 2 1 1
Process 4: 0 0 2
Enter the request matrix:
Process 0: 0 0 0
Process 1: 2 0 2
Process 2: 0 0 0
Process 3: 1 0 0
Process 4: 0 0 2
Enter the available resources: 0 0 0

System is in safe state.
Safe Sequence is: P0 P2 P3 P4 P1

Process returned 0 (0x0)   execution time : 73.878 s
Press any key to continue.
```